Global ROV Audit & Triangulation

Measuring RPKI Route Origin Validation Adoption

Through Dependency Graph Analysis

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A framework to answer not just "Is ROV enabled?" but "Is this network *protected*?"

The Problem: BGP Is Still Mostly Trust-Based

The Border Gateway Protocol underpins all global routing. It was designed for **cooperation**, not adversarial environments.

What can go wrong

- **Route Hijacking** — attacker announces your prefix, attracts your traffic
- **Route Leaks** — misconfiguration propagates invalid paths at global scale
- **Prefix Interception** — traffic silently redirected through a third country

Real incidents

- **2010**: China Telecom hijacked 15% of global routes for 18 minutes
- **2018**: BGP leak redirected Google traffic through Nigeria, Russia, China
- **2022**: Cloudflare, Amazon, AT&T routes hijacked via RPKI-invalid prefix
- **2024**: Multiple large-scale leaks from South Asia transit networks

> *BGP security is not an individual property. It is a **collective one**.*

RPKI: The Two-Part Solution

ROA — Route Origin Authorization

"I claim this prefix"

- Published in a signed certificate
- Links an ASN to the prefix(es) it is authorized to originate
- Stored in RPKI repositories (RIRs)
- Passive: does nothing by itself

Think of it as a signed deed of ownership

ROV — Route Origin Validation

"I verify what I receive"

- Routers check incoming BGP announcements against the RPKI
- If a route is **RPKI-Invalid** → drop it
- Must be configured and enforced at each AS
- Active: this is what actually stops attacks

Think of it as checking that deed at the border

Both must exist for the system to work. **ROA without ROV** = a signed deed that nobody checks. **ROV without ROA** = a border inspector with no database to query.

The Question Everyone Gets Wrong

The naive question

> *"Does AS12345 have ROV enabled?"*

Simple binary. Most existing tools stop here.

Problems:

- Doesn't account for inheritance
- Ignores upstream topology
- Misses "clean pipe" protection
- Can't distinguish active vs passive protection

The right question

> *"Is AS12345 protected from invalid route propagation — and how?"*

Requires dependency graph analysis.

Protection sources:

- **Active**: AS filters invalids itself
- **Passive**: upstream provider filters on its behalf
- **Partial**: some upstreams filter, some don't
- **None**: invalid routes flow freely

Philosophy

Herd Immunity for the Internet

Herd Immunity: The Core Concept

> *Just as epidemic disease can be suppressed when enough individuals are immune,* ***BGP hijacks can be suppressed when enough transit networks enforce ROV.***

How it applies to BGP

The internet has a power-law structure. A small number of **Tier 1 and large transit networks** carry a disproportionate share of global traffic.

If *those* networks enforce ROV, the invalid routes they drop **never reach the billions of end-users** downstream — regardless of those users' own AS configuration.

The cone model

Every network has a **customer cone**: the set of all ASes reachable downstream through it.

```
Tier 1 (cone = 70,000)
  └─ Regional ISP (cone = 500)
       └─ Local ISP (cone = 20)
            └─ Enterprise (cone = 1)
```

ROV at the Tier 1 protects *everyone* in its cone — ~70,001 networks.

What This Tool Measures

Traditional approach

Check if a network has ROV enabled. List networks. Count them.

Result: "X% of networks have ROV."

Misleading — a small stub AS and a Tier 1 backbone count equally.

Our approach

Weight by **customer cone size** (downstream reach). Trace the dependency graph for every network.

Result: "X% of global routable traffic flows through networks that will drop RPKI-invalid routes."

Meaningful — weighted by actual internet impact.

We also identify:

- Networks that are protected **passively** (clean pipe) even without local ROV
- Networks that protect others but **don't protect themselves** (Glass Houses)
- Networks doing everything right but whose effort is **nullified by their upstream** (Screaming Into the Void)

Architecture

How the Tool Works

System Overview: Four Phases

```
┌──────────────────────────────────────────────────────────────────┐
│  Phase 1                Phase 2                Phase 3         Phase 4│
│  BGP Topology  ──────►  Data Hydration  ──►  Audit    ──────► Reports│
│  (Go)                   (Python)               (Python)       (Python)│
└──────────────────────────────────────────────────────────────────┘
       │                       │                     │                │
  400MB MRT dump         6 data sources       Verdict logic    30+ reports
  Valley-free logic      Zero-scrape arch     120k ASNs         Country deep dives
  Cone calculation       Per-AS JSON cache    Dependency walk   Herd immunity calc
```

Data sources used:

Source	Data
RIPE RIS (MRT dumps)	Raw BGP routing tables — provider/customer relationships
APNIC Labs	ROA signing percentage per ASN
BGP.Tools	ASN names, ROV status tags
Cloudflare IsBGPSafeYet	Known-secure operators
IPtoASN	Geolocation (country codes)
RIPE Atlas	Active forensic probe results

Phase 1: Building the Internet's Topology Map

Input

RIPE RIS `latest-bview.gz` (~400MB)
MRT binary format BGP table dump

Process (Go)

1. Parse all BGP AS-PATH sequences
2. De-prepend repeated ASNs
3. Classify each hop as provider→customer or peer→peer using **Valley-Free** (Gao-Rexford) logic
4. Build `relationships.csv`
5. Run transitive closure → **customer cone sizes**

Output: final_as_rank.csv — 70,000+ ASNs ranked by customer cone size

Valley-Free Routing Logic

```
RULE: A packet can travel UP to a provider,
      ACROSS to a peer, and DOWN to a customer.
      It cannot go UP again after going DOWN.

  Tier1 ——— Tier1          ← peering
    |
  Transit                  ← provider→customer
    |
  Regional                 ← provider→customer
    |
  Stub                     ← customer (leaf)
```

Any path violating this model indicates a route leak or misconfiguration.

Phase 2: Data Triangulation — Zero-Scrape Architecture

We hydrate the topology with security metadata from **six independent sources**.

What we fetch

- **ASN Name** (BGP.Tools CSV bulk)
- **Country Code** (IPtoASN, triple-validated)
- **ROV Status** (BGP.Tools tags + Cloudflare)
- **ROA Signing %** (APNIC Labs JSON API)
- **Tier 1 classification** (curated list of 27 known backbone providers)
- **Upstream provider list** (from Phase 1 relationships)

Per-ASN JSON cache

```
{
  "asn": 10000,
  "name": "Example ISP",
  "is_tier1": false,
  "cone_size": 142,
  "upstreams": [3356, 1299, 6939],
  "cc": "US",
  "roa_signed_pct": 78.3,
  "roa_stats": {
    "valid": 47,
    "total": 60
  }
}
```

~120,000 files, one per ASN

Design principle: no scraping, no web parsing — only bulk APIs, published datasets, and structured endpoints.

Phase 3: The Audit — Verdict Logic

Each network is classified by walking its **upstream dependency graph**.

```
for each AS in database:
    if is_tier1:
        verdict = CORE:PROTECTED or CORE:UNPROTECTED  (based on ROV status)
    else:
        secure_feeds = count(upstreams where verdict == PROTECTED)
        total_feeds  = count(upstreams)

        if secure_feeds == total_feeds:   → SECURE (Full Coverage)
        if secure_feeds == 0:             → VULNERABLE (No Coverage)
        if 0 < secure_feeds < total_feeds:→ PARTIAL (Mixed Feeds)

        if cone_size < threshold:
            prefix verdict with STUB:
```

Key insight

Protection is **transitive**. A network is secure if *all* paths to the DFZ pass through an ROV-enforcing router — regardless of where in the path that enforcement happens.

Key caveat

Partial is dangerous. A network with 3 upstream providers — 2 secure, 1 leaking — is effectively vulnerable, because attackers will always exploit the leaking path.

The Verdict Taxonomy

Verdict	Meaning	Count	Traffic Impact
CORE: PROTECTED	Tier 1 with ROV enforced	22	**25.9%**
CORE: UNPROTECTED	Tier 1 without ROV	5	~0%
SECURE (Full Coverage)	All upstreams are secure	1,179	**52.1%**
SECURE (Active Local ROV)	Filters invalids locally	678	**3.5%**
PARTIAL (Mixed Feeds)	Some upstreams vulnerable	3,989	6.1%
VULNERABLE (No Coverage)	Upstream accepts invalids	55,859	0.3%
NOT ROUTED	No prefixes announced	36,986	—
Unverified (Transit/Peer?)	Unknown — needs forensics	187	12.1%

Total networks in database: 120,444

Fully Protected: 23,949 (19.9%) · Partially Protected: 3,989 (3.3%) · Vulnerable: 55,338 (45.9%)

Phase 4: Analysis Reports

From the master audit CSV, we generate targeted analytical reports:

Herd Immunity
What % of global traffic flows through ROV-enforcing networks?
Weighted by customer cone size.

ROV Quadrant Analysis
2×2 matrix: who filters vs. who signs. Identifies system failures.

ROA Signing Hygiene
Glass Houses: networks that filter others but don't sign their own routes.

Country Deep Dives
Per-nation security posture, critical ISPs, and regional recommendations. 30+ countries covered.

ASPA Readiness
Estimates transition cost and benefits of ASPA adoption across transit networks.

Forensic Verification
RIPE Atlas active probes to confirm or deny ROV enforcement in "Unverified" large networks.

Findings

What the Data Shows

The Core: 27 Tier 1 Networks

These 27 networks are the backbone of the global internet. Their combined customer cones span virtually every routable AS on earth.

Status as of March 2026

```
CORE: PROTECTED  (22 networks)
 ✓ AS6939   Hurricane Electric
 ✓ AS1299   Arelion (Telia Carrier)
 ✓ AS3356   Lumen (Level 3)
 ✓ AS174    Cogent Communications
 ✓ AS6461   Zayo Bandwidth
 ✓ AS9002   RETN Limited
 ...and 16 more

CORE: UNPROTECTED (5 networks)
 ✗ AS4134   China Telecom Backbone
 ✗ AS4837   China Unicom Backbone
 ✗ AS9808   China Mobile Backbone
 ✗ AS2828   Verizon Business
 ✗ AS9929   China Unicom Industrial
```

Why this matters

The 5 unprotected Tier 1 networks collectively provide transit to hundreds of millions of end-users.

An invalid route announced against a network they transit will propagate globally — because there is no ROV enforcement at the last backstop.

22/27 Tier 1s protected = 81% of the Core secured.

The 5 holdouts represent disproportionate geopolitical risk — concentrated in China (4 networks) and US (Verizon Business).

Herd Immunity: Current Status

Top 100 largest networks

```
Networks with ROV: 86 / 100  (86.0%)
Traffic protected: 85.3%
```

Top 1,000 transit networks

```
Networks with ROV: 489 / 1000  (48.9%)
Traffic protected: 82.5%
```

Interpretation

The herd immunity threshold is approximately 80% of traffic.

We are at **82.5%** by transit weight. The core of the internet has, in practical terms, **achieved herd immunity** for well-originated routes.

However, this is not uniformly distributed:

- Strong in Europe, North America, Oceania
- Significant gaps in South/Southeast Asia
- Structural gaps in China (Tier 1 holdouts)

> *CONCLUSION: The core is essentially safe. The tail risk is concentrated in specific geographic regions and specific large networks.*

The High-Impact Holdouts

If these networks enabled ROV, global protection would jump meaningfully.

Rank	ASN	CC	Cone	Network
#154	AS4134	CN	387	China Telecom Backbone
#177	AS45820	IN	319	Tata Teleservices ISP
#217	AS9730	IN	234	Bharti Telesonic Ltd
#342	AS17762	IN	129	Tata Teleservices Maharashtra Ltd
#364	AS4837	CN	111	China Unicom Backbone
#377	AS33132	US	104	Crown Castle Fiber LLC
#414	AS12357	ES	92	Vodafone España S.A.U.
#507	AS9808	CN	69	China Mobile Backbone
#828	AS5384	AE	33	Emirates Telecommunications

India (IN) also appears 4× in the top holdouts. South Asia is the primary geographic gap in global ROV adoption after China.

The ROV Quadrant Analysis

A 2×2 matrix reveals four structural failure modes in the system:

Q1: Gold Standard ✓

Provider filters invalids AND customers sign ROAs

```
AS6939  Hurricane Electric      60.1% signing
AS1299  Arelion (Telia)         65.8% signing
AS9002  RETN Limited            67.7% signing
```

System working as intended.

Q3: Glass Houses ⚠️

Provider filters, but own routes unsigned

```
AS3356  Lumen (Level 3)         5.0% signing
AS174   Cogent Communications   9.6% signing
AS3216  Vimpelcom               0.2% signing
```

They protect others. They are unprotected.

Q2: Screaming Into the Void ⚠️

Customers signed ROAs, but provider leaks

```
AS24482  SG.GS (Singapore)      74.0% signing
AS37721  VTS (Burkina Faso)     66.6% signing
```

Customers did their homework. Provider negated it.

Q4: The Swamp ✗

Vulnerable provider AND unsigned customers

```
AS38255  CERNET (China)          0.0% signing
AS9930   TIME dotCom (Malaysia) 52.8% signing
```

Complete failure on both axes.

Glass Houses: The Asymmetry Problem

Networks filtering invalids but not signing their own routes:

ASN	Cone	ROA Signed
AS3356 Lumen	65,326	5.0%
AS174 Cogent	64,687	9.6%
AS3216 Vimpelcom	38,066	0.2%
AS33891 Core-Backbone	33,466	0.0%
AS4766 Korea Telecom	1,045	3.6%

Why this is a problem

If Lumen (AS3356) were hijacked, the attacker would propagate routes to **65,326 downstream networks**.

Lumen's own ROV filters would detect the hijack — but the hijack is *of Lumen's own routes*, so there is nothing to filter. The attack succeeds against the very network doing the filtering.

ROV enforcer ≠ ROA signer. You need both.

Global ROA signing status:

Fully signed (>90%): 38,901 networks (32.3%) · Partially signed: 7,810 (6.5%) · Unsigned: 73,733 (61.2%)

Screaming Into the Void

Networks that have done everything right — 100% ROA signing — but whose upstream provider does not enforce ROV:

ASN	CC	Cone	Provider feeds leaking
AS45820 Tata Teleservices	IN	319	1/1
AS17762 Tata Maharashtra	IN	129	2/2
AS23688 Link3 Technologies	BD	44	1/1
AS141731 Max Hub	BD	41	3/3
AS4007 Subisu Cablenet	NP	32	1/1

The ROA signing tragedy

These networks published signed ROA records. Their routes are verifiable. Their prefixes can be validated by any ROV-capable router.

But their upstream provider does not perform that validation. Any invalid announcement against them will be accepted and propagated upstream — the ROA signatures are ignored.

ROA without upstream ROV is security theater.

The problem is not with these networks. It is with their transit providers.

Country-Level Analysis: South Asia Focus

The most significant regional gap. India and Bangladesh dominate the holdout list.

India

- Largest holdouts: Tata Teleservices, Bharti Telesonic
- Major transit networks without ROV serve ~500+ downstream ASNs
- Economic consolidation means a small number of large ISPs control most routes
- **Policy lever**: TRAI could mandate ROV for licensed telecom operators

Bangladesh

- 4 networks appear in "Screaming Into the Void" list
- Link3 Technologies (AS23688, cone 44) is fully signed but upstream leaks
- Rapid growth of ISP sector without routing security requirements

Nepal

- Subisu Cablenet (AS4007) — fully signed, upstream leaks
- Nepal Telecom (Tier 2) is partially signed
- Small market; depends almost entirely on Indian transit — inherits India's gaps

China (structural)

- 4 of 5 unprotected Tier 1 networks are Chinese operators
- Unique position: state-controlled networks
- Resolution likely requires regulatory/government-level engagement, not technical persuasion

Active Forensics: RIPE Atlas Verification

Some large networks appear as "Unverified" — they have ROV tags in BGP.Tools but no confirmed data.

The method

1. Identify target ASN with "Unverified" status
2. Find a RIPE Atlas probe inside that AS
3. Announce a **crafted RPKI-Invalid test prefix**
4. Run traceroutes from that probe to a controlled endpoint
5. Compare path for **Valid** vs **Invalid** prefix

If the Invalid trace drops at the network boundary → **ROV confirmed**.
If both traces reach the same endpoint → **leaking confirmed**.

What this produces

```
# verify_forensic_path_v2.py output:
Target ASN: AS12345
Probe: #987654 (inside AS12345)

Valid prefix trace:
  1. 10.x.x.x  (internal)
  2. 203.x.x.x (upstream)
  3. DESTINATION REACHED ✓

Invalid prefix trace:
  1. 10.x.x.x  (internal)
  2. REQUEST TIMED OUT
  ← route dropped at AS12345 border

VERDICT: ROV CONFIRMED. Reclassifying as SECURE.
```

Results are fed back into the main audit, improving classification accuracy for large "Unverified" networks (combined cone ~12% of global traffic).

The Stack

Technical Implementation

Technology Choices

Go — BGP Processing

The raw MRT dump is 400MB of binary data. Processing it in Python would take 20+ minutes.

Go processes it in **under 60 seconds** using concurrent goroutines for path parsing and valley-free classification.

```go
// Concurrent MRT processing
func processPaths(paths []BGPPath) {
    sem := make(chan struct{}, workers)
    for _, path := range paths {
        sem <- struct{}{}
        go func(p BGPPath) {
            defer func() { <-sem }()
            classifyRelationships(p)
        }(path)
    }
}
```

Python — Analysis & Reporting

All analysis, data triangulation, and report generation is in Python.

- `pandas` for CSV manipulation across 120k rows
- `asyncio` + `aiohttp` for parallel ROA stat fetching
- `ripe.atlas.cousteau` for forensic probe coordination

Zero-Scrape Architecture

No HTML parsing. No browser automation. Every data source used is:

- A bulk-downloadable CSV or compressed TSV
- A documented JSON API
- A structured public dataset

This makes the tool **reproducible and respectful** of data providers.

Data Pipeline: End to End

```
RIPE RIS                BGP.Tools             APNIC Labs
latest-bview.gz         rov.csv               rpki-history/
(400MB MRT)             (ASN names/tags)      (ROA signing %)
     │                       │                     │
     ▼                       │                     │
bgp-extractor (Go)           │                     │
  - Parse AS-PATHs           │                     │
  - Valley-free classify     │                     │
  - Output relationships.csv │                     │
     │                       │                     │
     ▼                       ▼                     ▼
cone-calculator (Go)    rov_no_scrape_v20.py (Python)
  - Transitive closure       - Fetch all metadata
  - Customer cone sizes      - Build data/parsed/as_*.json
  - final_as_rank.csv        - 120,000 per-ASN JSON files
          │                       │
          └───────────────────────┘
                      │
                      ▼
              rov_global_audit_v20.py
                 - Walk dependency graph
                 - Assign verdicts
                 - rov_audit_v20_final.csv
                      │
              ┌───────┼───────┐
              ▼       ▼       ▼
      herd_immunity   quadrant_v3   country_deep_dive
      .py             .py           .py [CC]
```

What This Means

Implications & Next Steps

Summary: The State of Internet Routing Security

The good news

- **86% of the top 100 networks** have ROV enabled
- **82.5% of transit-weighted traffic** passes through ROV-enforcing nodes
- Herd immunity threshold (~80%) has been **crossed** for well-originated routes
- ROA signing growing: 32% of networks now >90% signed

The remaining problems

1. **5 Tier 1 networks** still do not enforce ROV — 4 are Chinese state carriers
2. **South Asia** (India, Bangladesh) has the largest holdout clusters
3. **Glass Houses** — large networks protect others but don't sign their own routes
4. **Partial protection** (3,989 networks) is nearly as bad as no protection for attackers who can exploit leaking upstreams
5. **187 large networks** remain forensically unverified (~12% of traffic impact)

Recommendations

For network operators

1. **Enable ROV** — use RIPE's RPKI Validator or RIPEstat to check your status
2. **Sign your own routes with ROA** — even if you already filter invalids
3. **Require ROV from customers** — make it a peering/transit requirement
4. **Check your upstreams** — if they leak, your ROA signing is wasted

For IXPs and transit providers

- Make ROV a **peering policy requirement**
- Publish your ROV status in PeeringDB
- Prefer customers/peers with ROA signing

For regulators and policy makers

- **South Asia**: TRAI (India), BTRC (Bangladesh) should consider ROV mandates for licensed telecom operators
- **China**: Engagement with MIIT on Tier 1 ROV adoption is the highest-leverage intervention globally
- **RIPE NCC**: Consider making ROA signing a requirement for IP space allocation renewal

For the IETF community

- Continued work on **ASPA** (AS Provider Authorization) to extend protection from origin to path
- Better tooling for **partial protection analysis** — the binary "secure/vulnerable" framing is insufficient

ASPA: The Next Layer

Current RPKI/ROV protects against **origin hijacks** only.

ASPA (AS Provider Authorization) extends this to **path hijacks** — route leaks where the origin is legitimate but the path is not.

What ASPA adds

Each AS publishes a signed record of its legitimate upstream providers.

```
AS12345 authorizes providers:
  - AS3356 (Lumen)
  - AS1299 (Arelion)
```

Routers can then validate that the full AS-PATH in a BGP announcement is consistent with published provider relationships — not just the origin.

Full analysis: reports/analyze_aspa_realistic.md

ASPA readiness (from our analysis)

The valley-free relationship data we already collect is **exactly the data ASPA would use**.

Our `relationships.csv` and cone calculations are structurally compatible with ASPA's provider authorization model.

Estimated transition coverage if top 1,000 transit networks adopted ASPA: **~71% of global transit** would have path-level validation.

Data Availability & Reproducibility

Open source

- Full codebase on GitHub: `IEISI-ORG/rov_audit`
- Licensed under **CC BY-NC 4.0**
- Free to use for research, education, and non-commercial analysis
- Pull requests welcome

Reproducibility

The entire pipeline runs on public data. No proprietary sources, no scraping, no API keys required for the core audit.

```
# Full pipeline, one command:
make all

# Or step by step:
make topology
make metadata
make audit
make reports
```

Cite as

> ***"Global ROV Audit & Triangulation Tool"***
> *A framework for measuring Internet Routing Security via Dependency Analysis.*
> *IEISI-ORG, 2026.*
> `github.com/IEISI-ORG/rov_audit`

Data freshness

- BGP table: updated daily (RIPE RIS publishes every 8 hours)
- APNIC ROA stats: updated weekly
- Full audit run: ~2 hours on commodity hardware

Reports in this presentation reflect data from **March 2026**.

Thank You

Global ROV Audit & Triangulation Tool

Questions welcome.

Key numbers to remember

- **120,444** networks audited
- **82.5%** of transit-weighted traffic protected
- **86/100** top networks have ROV
- **5** Tier 1 networks still holdouts
- **3 of them** are Chinese state carriers

Resources

- Code + data: `github.com/IEISI-ORG/rov_audit`
- License: CC BY-NC 4.0
- Reports: 30+ country deep-dives, all public

"The internet's security is a collective property. No network is an island."